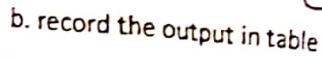
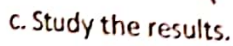


Name: _____

$\tau = 2 / T = 0.6, 1, 3 / A = 1$

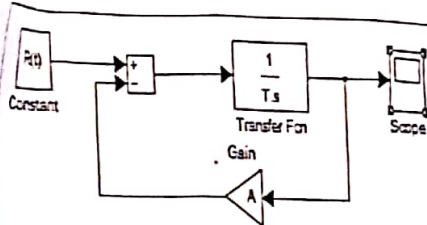


c. Plot the output

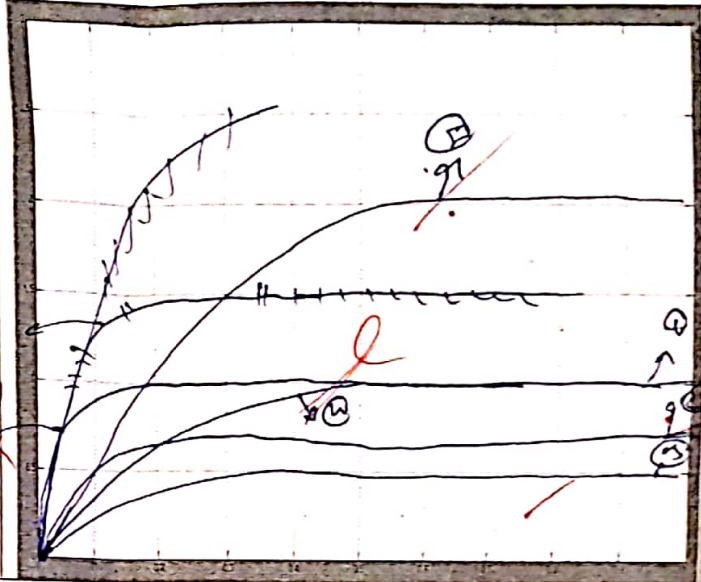


Feedback سے جبات

Q2. Plot the output response for each of the following conditions



1. $R(t)=1, T=1, A=0$
2. $R(t)=1, T=1, A=1$
3. $R(t)=1, T=1, A=3$
4. $R(t)=1, T=2, A=1$
5. $R(t)=2, T=1, A=1$



Q3. For a system given where $Y(t)$ is system output and $m(t)$ is system input : $r(t) = 2$

$$y'''(t) - 3y''(t) + 2y'(t) + 4y(t) = r(t)$$

a. Determine the state space matrices

Parameters

A:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 & -4 & -2 & 3 \end{bmatrix}$$

B:

$$\begin{bmatrix} 0 & 0 & 0 & 2 \end{bmatrix}$$

C:

$$\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

D:

$$\begin{bmatrix} 0 \end{bmatrix}$$

$$y'''(t) = 3y''(t) - 2y'(t) - 4y(t) + r(t)$$

$$x_1 = y \rightarrow x_1' = y' = x_2$$

$$x_2 = y' \rightarrow x_2' = y'' = x_3$$

$$x_3 = y'' \rightarrow x_3' = y''' = 3x_3 - 2x_2 - 4x_1 + 2$$

$$\begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -4 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} r(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \end{bmatrix} r(t)$$

b. Solve this equation by the transfer function

$$s^3 y(s) - 3s^2 y(s) + 2s y(s) + 4y(s) = r(s)$$

$$y(s)(s^3 - 3s^2 + 2s + 4) = r(s)$$

$$\frac{y(s)}{r(s)} = \frac{1}{s^3 - 3s^2 + 2s + 4}$$